MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 8354

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen, ceramic-metal

See figure 1

Mounting position: Any

Weight: 2 pounds (907.2 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	egy	egx	Ecc	ib	Ib
Unit:	Vac	kv	kv	Vdc	v	v	Vdc	a	Adc
Maximum:	6.8	25	25 (Note 1)	---	1,500 (Note 2)	400	-200	1,000	2.2
Minimum:	5.8	---	5% epy	1,500	500	---	---	---	---
TEST CONDITIONS: (Note 8)	6.3	25	---	---	450	---	---	---	---

ABSOLUTE RATINGS:

Parameter:	Ip	tj	prr	dik/dt	Pb	tk	TA	Cooling	Eres
Unit:	Aac	us	---	a/us	---	sec	°C	---	Vac
Maximum:	40	0.005	4,000	5,000	25×10^9 (Note 7)	---	+85	--- (Note 7)	6.8
Minimum:	---	---	---	---	---	300	-55	---	5.8
TEST CONDITIONS: (Note 8)	---	---	---	---	---	300	Ambient	---	6.3

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
Ⓒ 3251	Pulse emission	-	ik = 1,500 a; tp = 5 µs ±10%; prr = 60 ±10%; tr = 0.5 µs (max); specified time interval = 2.5 µs	egk	---	250	v
3241	Heater current (cathode)	-		If	11	18	Aac
3241	Heater current (reservior)	-		Ires	3	8	Aac
Ⓒ 3246	Operation (1)	3,4,5	Ef = Eres = 5.8 Vac; epy = 24 kv; Ib = 2.2 Adc; t = 1 hour	egy	---	450	v
Ⓒ 3246	Operation (2)	10	Operation (1), except Ef = Eres = 6.3 Vac; t = 5 hours	egy	---	450	v
Ⓒ 3267	Instantaneous starting	3,6	epy = 18 kv (min); Ef = Eres = 6.8 Vac	---	---	---	---
Ⓒ 3247	DC anode voltage for conduction	-	Ef = Eres = 5.8 Vac	Ebb	---	1,000	Vdc
	Quality conformance inspection, part 2						
Ⓒ 3256	Anode delay time	-	Operation (1), except t = 120 seconds	tad	---	0.5	µs
Ⓒ 3256	Anode delay time drift	9	Anode delay time	Δtad	---	0.15	µs
Ⓒ 3261	Time jitter	12	Operation (2); tk = 300 seconds	tj	---	0.005	µs
	Quality conformance inspection, part 3						
1031	Sweep-frequency vibration	11	No voltages applied	---	---	---	---
1041	Shock	11	250 G; no voltages applied	---	---	---	---
---	Shock and sweep-frequency vibration end points	-	DC anode voltage for conduction	Ebb	---	1,250	Vdc
---	Life test	3	Group C; t = 500 hours	---	---	---	---
---	Life test end points:						
3246	Operation (1) and (2)	-	Except note 5	egy	---	450	v
3247	DC anode voltage	13		Ebb	---	1,250	Vdc
3261	Time jitter	-		tj	---	0.005	µs

NOTES:

1. In pulsed operation, the peak inverse voltage exclusive of a spike of 0.05 µs maximum duration, shall not exceed 5 kv during the first 25 µs following the anode pulse.

2. The driver pulse, measured at the tube socket with the thyratron grid disconnected: tr = 0.35 µs maximum; grid pulse duration = 2 µs minimum. The impedance of the drive circuit shall be 50 to 200 ohms.

3. The anode circuit constants shall be chosen for a resonant charging condition of prr = 4,000 minimum. At epy = 24 kv and ib = 580 a minimum, the rate of rise of the current pulse (dik/dt) shall be 5,000 a/µs minimum. The pulse width (tp) shall be 2.4 µs ±10 percent. The grid pulse measurements at the tube socket with the thyratron grid disconnected shall be as follows: tr = 0.35 µs minimum, tp = 2.0 µs maximum, Zg = 200 ohms minimum, Ecc = -70 V [illegible].

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NOTES: -Continued

4. The anode temperature and grid seal temperature shall not exceed 350°C at any time during this test.

5. Anode voltage shall be applied as specified in note 6. Full anode voltage (epy = 24 kv) shall be reached within 5 seconds after starting. The tube shall operate at the specified anode voltage for a total of 6 hours which may include two interruptions during the initial 30 minutes and one interruption during the last 5-1/2 hours.

6. The tube shall operate satisfactorily on push button starting within two attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 18 kv minimum within 0.03 second. (The filter in the rectifier shall be designed so that epy reaches at least 9 kv within 0.015 second). Any tube failing to start within three attempts will be considered a failure.

7. There shall be no artificial anode or envelope cooling directed onto the tube. Blowers or anode radiators shall not be used.

8. The heat dissipator, Hughes Aircraft Co. Part No. 529033 (grid radiator), or equivalent shall be used during all tests of this specification.

9. This test shall be performed simultaneously with operation (1) test. Anode delay time measurement shall be made at the end of 2, 4, and 30 minutes of the operation (1) tests. The anode delay time drift (Δtad) measurement is the numerical difference between 2 minute and 4 minute, or 2 minute and 30 minute anode delay time readings, whichever is greater.

10. Operation (2) shall be a continuation of operation (1) with the specified change in operating conditions made after 1 hour.

11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

12. The tube shall be tested by applying a peak forward anode voltage, as specified in the test conditions for the time jitter test immediately after the tk. The tj shall be not greater than the amount specified after 120 seconds of operation.

13. The tube shall operate at the specified anode voltage for a total of 6 hours which may include three interruptions during the initial 30 minutes and three interruptions during the last 5-1/2 hours.

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N036)

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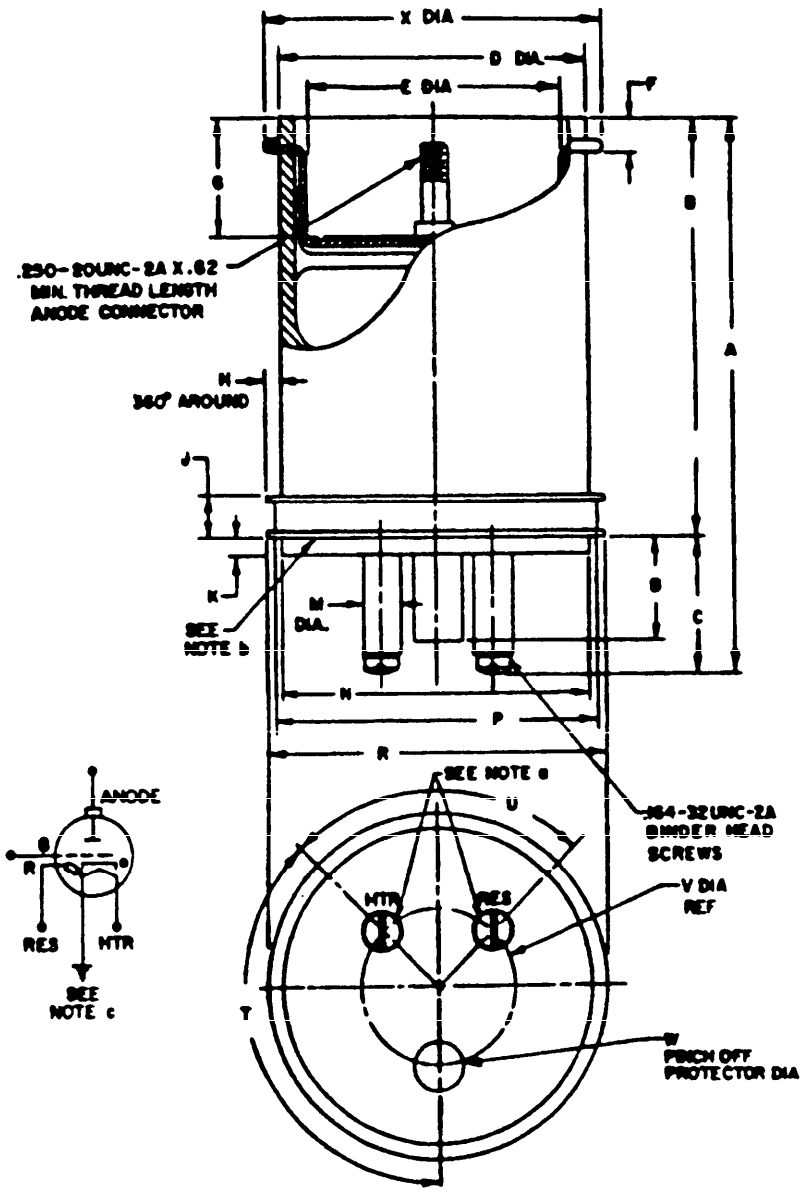


FIGURE 1. Outline dimensions of tube type 8354.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
K	.100 (2.54)	.180 (4.57)
N	2.985 (75.82)	3.015 (76.58)
Quality conformance inspection, part 2		
A		5.895 (149.73)
B	3.945 (100.20)	4.340 (110.24)
C		1.555 (39.50)
D	2.985 (75.82)	3.015 (76.58)
E	2.410 (61.21)	2.530 (64.26)
F		.313 (7.95)
G	.980 (24.89)	1.355 (34.42)
H	.093 (2.36)	
J	.520 (13.21)	.570 (14.48)
M		.375 (9.53)
P	3.235 (82.17)	3.265 (82.93)
R	3.245 (82.42)	3.305 (83.95)
S		1.081 (27.46)
W		.469 (11.91)
X		3.430 (87.12)
Reference dimensions		
T	135°	
U	90°	
V	1.500 (38.10)	

NOTES:
a. Reservoir and heater terminals shall be identified and marked on tube base as specified herein.
b. Cathode ring connection.
c. Other side of reservoir and heater terminals are internally connected and are common to cathode.

FIGURE 1. Outline dimensions of tube type 8354 - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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